PROJECT REPORT

AUTO TICKET CLASSIFICATION FLOW DESIGNER

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE COURSE

Submitted by

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PIRAIKUDIYERUPPU.

ACADEMIC YEAR: 2026–2027

ACKNOWLEDGEMENT

We express our sincere gratitude to the Management of **Sivanthi Arts and Science College for Women, Piraikudiyeruppu** for providing us with the opportunity and facilities to complete this project successfully.

We extend our heartfelt thanks to our respected **Head of the Department and Project Guide, Jency Diana MSC, MPhil, MED..,** Department of Computer Science, for their valuable guidance, continuous support, encouragement, and suggestions throughout the development of this project.

We also thank all the faculty members of the Department of Computer Science for their support and cooperation during the completion of this project.

Finally, we express our sincere thanks to our parents and friends for their encouragement and support in completing this project successfully.

ABSTRACT

The project “Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer” demonstrates the use of workflow automation in the IT procurement process. Organizations regularly receive requests for laptops and other IT equipment. When these requests are handled through disconnected manual activities, request processing can become difficult to

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LIST OF ABBREVATIONS

Abbreviation	Expansion
IT	Information Technology
ITSM	IT Service Management
UI	User Interface

1. INTRODUCTION

1.1 Background of the Project

In modern educational institutions and organizations, Information Technology plays an important role in supporting daily activities. Users may face different technical problems related to network connectivity, hardware devices, system access, applications, and system performance. When such problems occur, users generally create IT support tickets to report their issues to the support team.

In a basic ticket management system, support staff manually read each ticket and identify the appropriate category and subcategory. For example, a network-related problem may be classified under the Network category, while a forgotten password may be classified under the Access category. When the number of tickets increases, manually performing these activities can consume considerable time and may lead to inconsistent classification.

The proposed Auto Ticket Classification project focuses on improving this process through automation. The project uses ServiceNow Flow Designer to create a workflow that can process newly created tickets and perform predefined actions. The system also uses structured fields and dependent choices to maintain consistent ticket information.

1.2 Project Overview

The Auto Ticket Classification system is developed using the ServiceNow platform. A custom table named School IT Ticket is created to store the required information about IT support requests.

The system contains important fields such as:

Field	Purpose
Number	Provides a unique ticket number
Caller	Identifies the person who raised the ticket
Category	Identifies the general type of issue
Subcategory	Provides more specific issue classification
Short Description	Provides a brief description of the issue
Description	Provides detailed information about the issue

The system uses Flow Designer to automate ticket-related activities. When a ticket is created, the configured flow can evaluate the ticket information and perform the required actions.

1.3 Problem Statement

Manual ticket classification can create several challenges in an IT support environment. Support staff need to review every ticket individually and select the correct category and subcategory. This can become difficult when many tickets are received within a short period.

The major problems identified are:

- ❿ Manual classification requires additional time.
- ❿ Similar issues may be classified differently.
- ❿ Incorrect category selection can affect ticket tracking.
- ❿ Support staff may have repetitive work.
- ❿ Unstructured ticket information can make reporting difficult.
- ❿ Users may select unrelated category and subcategory values.
- ❿ Manual notification can create additional work.

Therefore, an automated and standardized ticket classification process is required to improve the efficiency of ticket management.

1.4 Aim of the Project

The main aim of this project is to develop an automated IT ticket classification workflow using ServiceNow Flow Designer.

The system is designed to provide a structured method for capturing IT support requests and automating the classification and notification process. By using predefined categories, subcategories, and workflow conditions, the system can reduce repetitive manual activities.The project also aims to provide a scalable foundation where new categories, subcategories, conditions, and actions can be added when required.By using predefined categories, subcategories, and workflow conditions, the system can reduce repetitive manual activities.The project also aims to provide a scalable foundation where new categories, subcategories, conditions, and actions can be added when required.

1.5 Objectives of the Project

The major objectives of the project are:

1. To create a structured School IT Ticket table.
2. To configure all required ticket fields.

3. To provide predefined categories for IT issues.
4. To configure relevant subcategories.
5. To implement dependent choices between category and subcategory.
6. To automate ticket processing using Flow Designer.
7. To provide automated email notification.
8. To reduce manual classification work.
9. To improve consistency in ticket data.
10. To create a maintainable and scalable workflow.

1.6 Scope of the Project

The project mainly focuses on the classification and automation of IT support tickets. It includes the creation of a custom ticket table, field configuration, category and subcategory configuration, dependent choices, workflow automation, and email notification.

The system can be used for common IT support problems such as:

- Network issues
- Hardware problems
- Access-related problems
- Performance-related problems
- Password-related issues
- Wi-Fi-related issues
- Projector-related issues
- Slow computer issues

The project can be further expanded with automatic assignment, priority calculation, escalation, dashboards, reports, and AI-based classification.

2. BRAINSTORMING AND IDEATION

2.1 Identification of the Problem

The brainstorming process began by identifying common problems faced in IT ticket management. The team considered how users report technical issues and how support staff process

those requests.During this stage, manual ticket classification was identified as a repetitive activity. Every new ticket requires someone to understand the issue and select the appropriate category and subcategory.

2.2 Understanding the Existing Process

The existing process generally involves the following steps:

1. User identifies a technical problem.
2. User creates an IT support ticket.
3. Support staff reviews the ticket.
4. The issue is manually classified.
5. Appropriate category and subcategory are selected.
6. The ticket is processed by the support team.
7. User may receive confirmation or further communication.

Although this process is simple, it becomes less efficient when the number of tickets increases.

2.3 Generation of Ideas

Different possible solutions were considered during brainstorming.

Idea	Purpose
Predefined choices	Reduce incorrect data entry
Dependent choices	Connect category with relevant subcategory
Automated workflow	Reduce manual processing
Email notification	Automatically inform the caller
Rule-based classification	Apply predefined classification logic
AI-based classification	Future advanced classification

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2.4 Selection of the Proposed Solution

ServiceNow was selected because it provides features for creating tables, fields, choice lists, workflows, notifications, and automation within the same platform.

Flow Designer provides a visual interface for designing workflows using triggers, conditions, and actions. This makes it suitable for a project where automation is an important requirement.The selected solution also provides flexibility for future development.

2.5 Expected Benefits

The proposed idea is expected to provide several benefits:

- Reduction in repetitive manual activities.
- Better consistency in ticket classification.
- Improved quality of ticket information.
- Faster processing of newly created tickets.
- Easier maintenance of classification rules.
- Better communication through email notifications.
- Possibility of future expansion.

2.6 Finalized Project Idea

After the brainstorming and ideation process, the final project was defined as:“Auto Ticket Classification using ServiceNow Flow Designer.”The project combines structured ticket data with workflow automation. The system is designed to classify and process IT tickets according to predefined business requirements while maintaining standardized information.

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3. REQUIREMENT ANALYSIS

3.1 Functional Requirements

Functional requirements describe the activities that the system must perform.The system should allow users to create an IT support ticket and enter the required information. It should store the ticket details in the custom School IT Ticket table.The system should also support category and subcategory selection, dependent choices, automated workflow execution, and email notification.

Major Functional Requirements

- ❿ Create a new IT ticket.
- ❿ Generate a unique ticket number.
- ❿ Select the caller.
- ❿ Select a suitable category.
- ❿ Select a related subcategory.
- ❿ Enter short and detailed descriptions.
- ❿ Process the ticket through Flow Designer.
- ❿ Send automated notification.
- ❿ Store the ticket information.

3.2 Non-Functional Requirements

Non-functional requirements describe the quality and performance expectations of the system.

Requirement	Description
Usability	System should be simple to use
Reliability	Workflow should execute consistently
Maintainability	Configuration should be easy to modify
Scalability	System should support future expansion
Consistency	Ticket data should follow standard values
Security	Access should be controlled according to platform permissions

3.3 Hardware Requirements

The project does not require specialized hardware. A standard computer or laptop with sufficient memory and processing capability is enough to access the ServiceNow development environment.The project does not require specialized hardware. A standard computer or laptop with sufficient memory and processing capability is enough to access the ServiceNow development environment.The project does not require specialized hardware. A standard computer or laptop with sufficient memory and processing capability is enough to access the ServiceNow development environment.The project does not require specialized hardware. A standard computer or laptop with

sufficient memory and processing capability is enough to access the ServiceNow development environment.

Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM recommended
- Keyboard and mouse
- Display monitor
- Stable internet connection

Since ServiceNow is accessed through a web browser, the system mainly depends on internet connectivity rather than local server hardware.

3.4 Software Requirements

The following software and platform components are required:

Software / Platform	Purpose
ServiceNow	Main development platform
Flow Designer	Workflow automation
Web Browser	Accessing ServiceNow
ServiceNow Developer Instance	Development and testing
Internet Connection	Access to cloud platform

3.5 Data Requirements

The system requires structured data to create and classify tickets.The data should be entered using appropriate field types and predefined choices wherever possible.The project does not require specialized hardware. A standard computer or laptop with sufficient memory and processing capability is enough to access the ServiceNow development environment.

The major data elements include:

- Caller information
- Ticket number
- Category
- Subcategory

- ➉ Short description
- ➉ Detailed description

\ The data should be entered using appropriate field types and predefined choices wherever possible.The project does not require specialized hardware. A standard computer or laptop with sufficient memory and processing capability is enough to access the ServiceNow development environment.

3.6 User Requirements

The system has two major types of users: ticket users and administrators/support staff.Ticket users should be able to create tickets and provide information about their technical problems.Administrators should be able to manage the table, fields, choices, dependencies, and workflow.The system should provide a simple interface so that users can enter the required information without confusion.

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4. PROJECT DESIGN PHASE

4.1 System Architecture Design

The system is designed around the ServiceNow platform. The custom School IT Ticket table acts as the central component for storing ticket information.

The overall process can be represented as:

User → Ticket Creation → School IT Ticket Table → Flow Designer → Classification/Actions → Email Notification

The workflow connects the ticket information with automated actions.

4.2 Table Design

A custom table named School IT Ticket is created for the project.

Table Details

Property	Value
Label	School IT Ticket
Table Name	u_school_it_ticket
Application	Global
Purpose	Store IT support tickets

The table is designed to store structured information about technical issues reported by users.

4.3 Field Design

The following fields are included in the table:

Field Label	Type	Purpose
Number	Auto Number	Unique ticket identification
Caller	Reference	Identifies ticket creator
Category	Choice	General issue classification
Subcategory	Choice	Detailed classification
Short Description	String	Brief issue description
Description	String / appropriate text field	Detailed issue information

Each field is selected based on the type of data that needs to be stored.

4.4 Category and Subcategory Design

The Category field contains predefined values.

Category	Example Subcategories
Network	Wi-Fi
Hardware	Projector
Access	Forgot Password
Performance	Slow Computer

The subcategory is dependent on the selected category. This ensures that users only select relevant subcategories.

For example, when **Access** is selected, the user should be able to select **Forgot Password**, rather than an unrelated network issue.

4.5 Flow Design

The workflow is designed using ServiceNow Flow Designer.The basic flow structure includes:

Trigger → Condition → Classification/Action → Notification → End

The trigger starts the flow when a ticket is created or when the specified condition occurs. Conditions are then evaluated, and appropriate actions are executed.The trigger starts the flow when a ticket is created or when the specified condition occurs. Conditions are then evaluated, and appropriate actions are executed.

4.6 Notification Design

The project includes automated email notification as part of the workflow.After a ticket is created, the system can send an email to the caller. The notification may include:

- Ticket number
- Caller information
- Short description
- Category
- Subcategory
- Ticket creation confirmation

5. PROJECT PLANNING PHASE

5.1 Project Planning Overview

Project planning defines the activities required to complete the project successfully. The project is divided into multiple stages so that development can be carried out systematically.The major stages include requirement analysis, design, configuration, development, testing, and documentation.

5.2 Task Identification

The following major tasks were identified:

No.	Task
1	Understand the business use case
2	Analyze requirements

3 Design the ticket structure

4 Create the custom table

5 Configure fields

6 Configure choices

7 Configure dependencies

8 Create Flow Designer workflow

9 Configure notification

10 Test the system

11 Prepare documentation

5.3 Resource Planning

The project requires both technical and human resources.

Technical Resources

- Computer/laptop
- Internet connection
- Web browser
- ServiceNow instance

Knowledge Resources

- Basic ServiceNow knowledge
- Flow Designer concepts
- Table and field configuration
- Workflow concepts
- Testing and documentation skills

5.4 Time Planning

The project activities are planned in a sequential manner.

Phase	Major Activity
Phase 1	Requirement Analysis
Phase 2	Brainstorming and Ideation

Phase 3 System Design

Phase 4 Table and Field Development

Phase 5 Flow Designer Development

Phase 6 Testing

Phase 7 Documentation

This planning helps ensure that the development activities are completed in an organized way.

5.5 Risk Planning

Several risks may occur during development.

Risk	Possible Solution
Incorrect field type	Verify field requirements
Dependency not working	Check dependent choice configuration
Flow not triggering	Verify trigger conditions
Incorrect classification	Review flow conditions
Email not sent	Verify notification configuration
Configuration error	Test individual components

Identifying these risks before implementation helps reduce development problems.

5.6 Project Completion Planning

The final stage includes complete testing, correction of identified errors, preparation of screenshots, documentation of configuration details, and preparation of the final project report.The project is considered complete when all major requirements are implemented and the system produces the expected results during testing.

6. PROJECT DEVELOPMENT PHASE

6.1 ServiceNow Environment Setup

Development begins by accessing the ServiceNow development environment. The required application and development area are selected before creating the project components.The development environment provides access to tables, fields, forms, Flow Designer, and other configuration tools required for the project.Development begins by accessing the ServiceNow development environment. The required application and development area are selected before creating the project components.The development environment provides access to tables, fields, forms, Flow Designer, and other configuration tools required for the project.

6.2 Custom Table Creation

The School IT Ticket custom table is created to store all project-related ticket records.The table acts as the main data structure of the application. Each ticket created by a user is stored as a record in this table.The table provides a structured foundation for further configuration and automation.

6.3 Field Configuration

After creating the table, the required fields are added.The Number field is configured to automatically generate ticket numbers. Development begins by accessing the ServiceNow development environment. The required application and development area are selected before creating the project components.The development environment provides access to tables, fields, forms, Flow Designer, and other configuration tools required for the project..

6.4 Choice and Dependency Configuration

The category choices are created based on the project requirements.

Example:

- Network
- Hardware
- Access
- Performance

Relevant subcategories are then created and connected to their corresponding categories.fter creating the table, the required fields are added.The Number field is configured to automatically generate ticket numbers. Caller is configured as a reference field so that a valid user can be selected.Category and Subcategory are configured as choice fields. Short Description and Description are used to capture information about the reported problem.This dependency improves data quality by preventing unrelated selections.

6.5 Flow Designer Implementation

The Flow Designer workflow is created after the table and fields are configured.The flow contains a trigger that determines when the automation should start. Conditions are then configured to evaluate the ticket information.Based on the configured conditions, the flow performs the required actions. The visual design of Flow Designer makes it easier to understand the sequence of automated activities.

The Flow Designer workflow is created after the table and fields are configured.The flow contains a trigger that determines when the automation should start. Conditions are then configured to evaluate the ticket information.Based on the configured conditions, the flow performs the required actions.

The visual design of Flow Designer makes it easier to understand the sequence of automated activities.6.6 Email Notification ConfigurationThe email notification is configured as part of the automated workflow.When the required event occurs, the flow performs the notification action and sends relevant ticket information to the caller.

This provides immediate confirmation and improves communication between the system and the user.The email notification is configured as part of the automated workflow.When the required event occurs, the flow performs the notification action and sends relevant ticket information to the caller.This provides immediate confirmation and improves communication between the system and the user.

7. PROJECT TESTING

7.1 Testing Objectives

Testing is performed to verify that the developed system works according to the project requirements.

The main objectives are:

- Verify table functionality.
- Verify field configuration.
- Verify choice values.
- Verify dependent choices.
- Verify Flow Designer execution.
- Verify classification logic.
- Verify email notification.
- Identify and correct configuration errors.

7.2 Test Case Design

Test cases are prepared for different system functions.

Test Case	Input/Action	Expected Result
TC01	Create new ticket	Ticket should be created
TC02	Enter caller	Caller should be stored
TC03	Select category	Category should be displayed
TC04	Select category	Related subcategory should appear
TC05	Create ticket	Flow should execute
TC06	Trigger notification	Email should be generated
TC07	Enter description	Description should be stored

7.3 Table and Field Testing

The custom table is tested by creating sample records. Each field is checked to ensure that the correct information can be entered and stored.The automatic Number field is also checked to ensure that a unique ticket number is generated for each record.

7.4 Dependency Testing

Dependency testing is performed by selecting different categories and checking the available subcategories.

For example:

Selected Category	Expected Subcategory
Network	Wi-Fi
Hardware	Projector
Access	Forgot Password
Performance	Slow Computer

The test is successful when only the relevant subcategories are displayed for each category.

7.5 Flow Testing

Flow testing is performed by creating sample tickets and checking whether the configured workflow executes correctly.The flow execution details are reviewed to identify whether the trigger, conditions, and actions are functioning as expected.If the flow does not execute correctly, the configuration is reviewed and necessary changes are made.

7.6 Notification Testing

Notification testing verifies whether the configured email is generated after the required ticket event.The following details are checked:

- ➉ Correct recipient
- ➉ Ticket number
- ➉ Ticket information
- ➉ Notification content
- ➉ Successful email generation

The test confirms that the notification process works according to the defined requirement.Notification testing verifies whether the configured email is generated after the required ticket event.

8. PROJECT DOCUMENTATION

8.1 Purpose of Documentation

Documentation provides a detailed record of the complete project development process. It explains the requirements, design, configuration, implementation, testing, and results.

Proper documentation also makes the project easier to understand, maintain, and modify in the future. It explains the requirements, design, configuration, implementation, testing, and results.

Proper documentation also makes the project easier to understand, maintain, and modify in the future.

8.2 Table Documentation

The custom School IT Ticket table is documented with its table name, fields, field types, and purposes.This documentation helps developers and administrators understand how ticket information is stored.The custom School IT Ticket table is documented with its table name, fields, field types, and purposes.This documentation helps developers and administrators understand how ticket information is stored.A field documentation table can be maintained as follows:

Field	Type	Purpose
Number	Auto Number	Ticket identification
Caller	Reference	User identification
Category	Choice	Main classification
Subcategory	Choice	Detailed classification
Short Description	String	Brief issue
Description	Text/String	Detailed issue

8.3 Flow Documentation

The Flow Designer workflow is documented with its trigger, conditions, actions, and notification steps.A simplified flow sequence is:

Ticket Created → Check Conditions → Classify Ticket → Perform Action → Send Notification

Screenshots of the Flow Designer configuration can be added to the report to provide evidence of implementation.The Flow Designer workflow is documented with its trigger, conditions, actions, and notification steps. This documentation helps developers and administrators understand how ticket information is stored

8.4 Testing Documentation

All important test cases and their results are documented.

The testing section includes:

- ➓ Test case number
- ➓ Test input
- ➓ Expected output
- ➓ Actual output
- ➓ Test status

This provides evidence that the project was tested systematically.

8.5 Screenshots and Evidence

Screenshots are an important part of the project documentation. They demonstrate the actual implementation of the project. Recommended screenshots include:

1. ServiceNow development environment
2. School IT Ticket table
3. Field configuration
4. Category choices
5. Subcategory choices
6. Dependent choice configuration
7. Flow Designer workflow

8. Created ticket
9. Flow execution
10. Email notification

8.6 Final Documentation

The final documentation combines all project information into a structured report. It includes the project introduction, brainstorming, requirements, design, planning, development, testing, results, and conclusion.The completed documentation serves as a reference for academic evaluation as well as future enhancement of the project.

9. CONCLUSION

9.1 Project Summary

The Auto Ticket Classification project was developed to improve the process of handling IT support tickets. The project uses ServiceNow to create a structured ticket management environment.A custom School IT Ticket table was created to store ticket information, while predefined categories and subcategories provide standardized classification.

9.2 Implementation Summary

The project implementation includes several major components:

- Custom ticket table
- Required ticket fields
- Category choices
- Subcategory choices
- Dependent choice configuration
- Flow Designer workflow
- Automated email notification
- Testing and validation

These components work together to create a basic automated ticket classification system.

9.3 Achievement of Objectives

The project successfully addresses the major objectives identified during the requirement analysis stage. The system provides structured ticket data, predefined classification values,

dependent choices, and workflow automation. These features help reduce repetitive manual activities and improve consistency.

9.4 Benefits of the System

The major benefits of the system include:

- Reduced manual effort
- Standardized ticket information
- Improved classification consistency
- Better ticket organization
- Automated workflow execution
- Automated user notification
- Easier maintenance
- Possibility for future expansion

9.5 Future Enhancements

Although the current project provides basic automated ticket classification, several advanced features can be added in the future. Possible enhancements include:

- AI-based ticket classification
- Automatic assignment to support groups
- Automatic priority calculation
- SLA-based escalation
- Ticket dashboards
- Reports and analytics
- Knowledge base integration
- Chatbot-based ticket creation
- Advanced email notifications
- Integration with external systems

AI or machine learning can be used in future versions to understand the natural language used in ticket descriptions and automatically determine the most suitable category and subcategory.

9.6 Final Conclusion

The **Auto Ticket Classification using ServiceNow Flow Designer** project demonstrates how workflow automation can be used to improve IT ticket management. The system provides a structured method for capturing technical issues and automating important ticket-related activities. By combining custom tables, structured fields, dependent choices, workflow automation, and email notification, the project reduces repetitive manual work and improves consistency in ticket information. The project also provides a strong foundation for future enhancements.

10. REFERENCES

10.1 Service Now Documentation

Service Now documentation can be referred to for understanding tables, fields, forms, Flow Designer, notifications, and other platform features used during the development of the project.

10.2 ServiceNow Developer Resources

ServiceNow Developer resources provide technical information and practical guidance for creating applications, configuring tables, managing fields, and developing workflows.

10.3 ServiceNow Flow Designer Resources

Flow Designer learning resources are used to understand triggers, conditions, actions, data handling, and workflow automation.

10.4 ServiceNow Learning Resources

ServiceNow learning modules provide information about IT Service Management concepts and the different features available in the ServiceNow platform.

10.5 Technical Articles and Tutorials

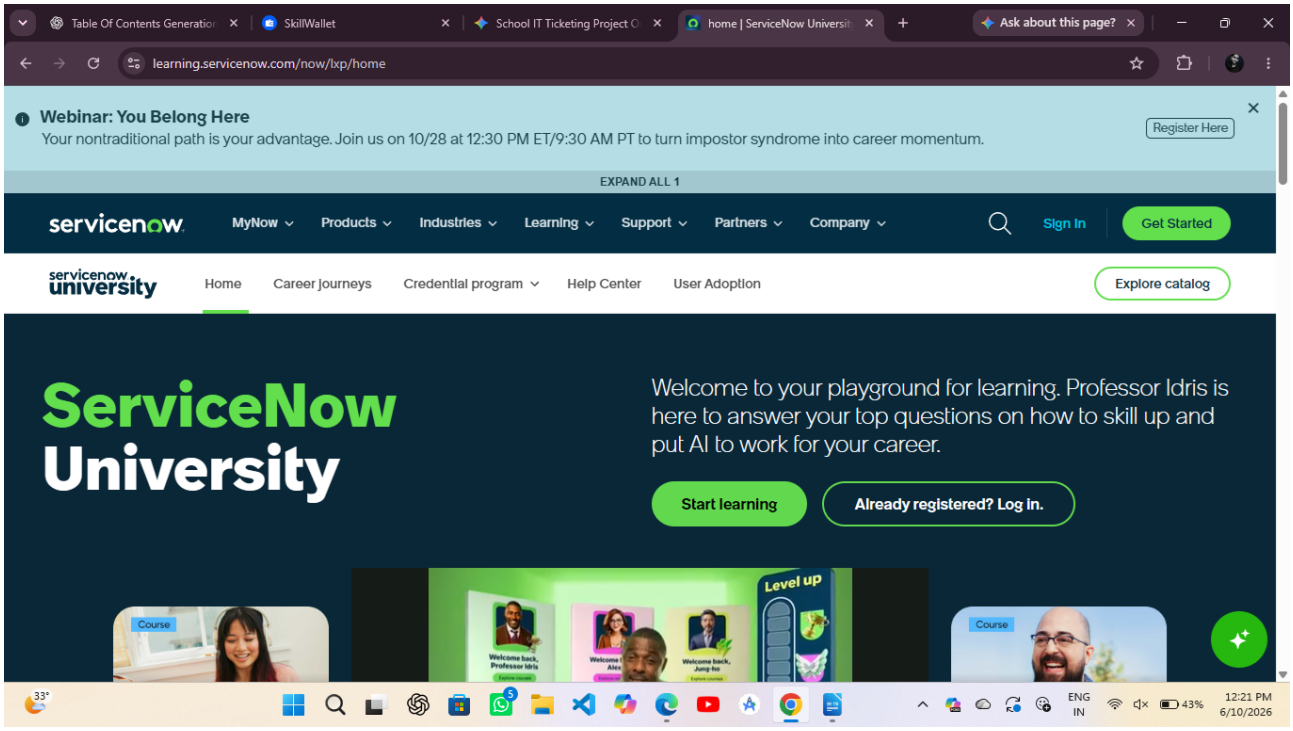
Technical articles and tutorials related to ServiceNow automation, ticket management, Flow Designer, and dependent choices can be used as supporting learning resources during project development.

10.6 Project Configuration and Testing Records

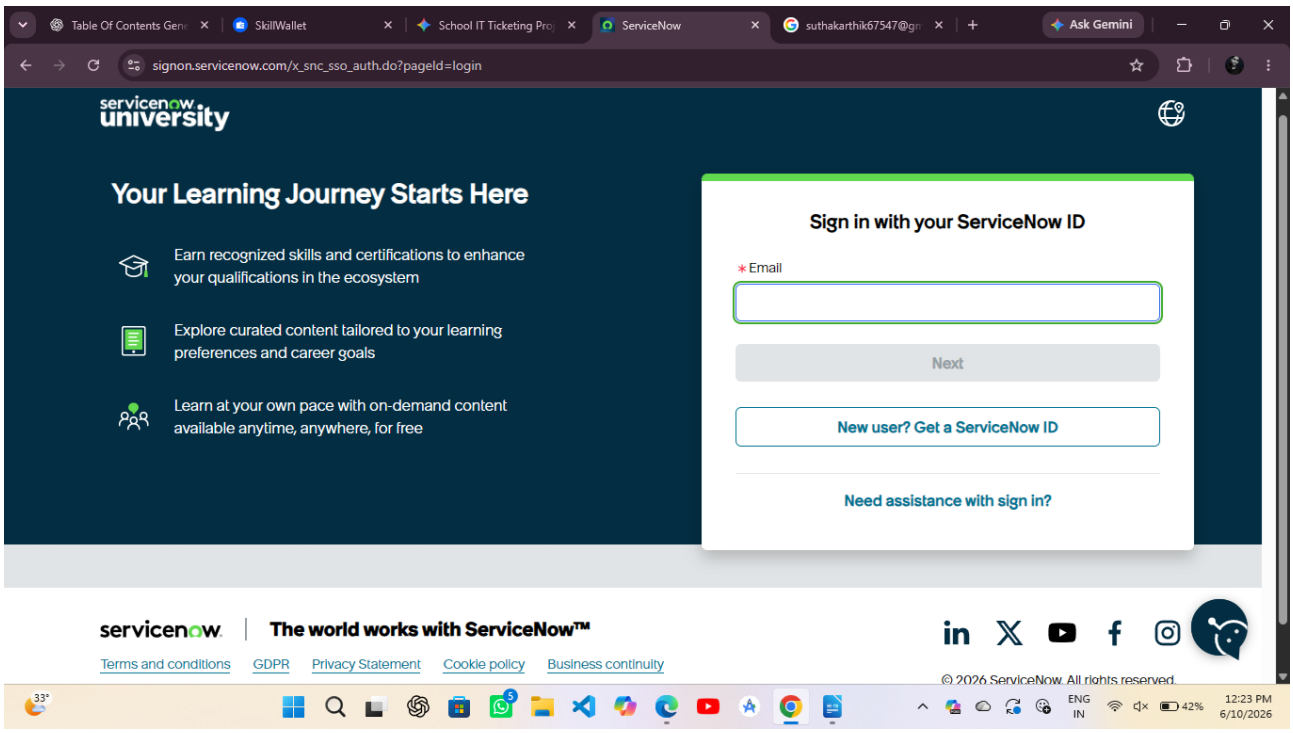
The project configuration details, screenshots, test cases, execution results, and observations collected during development are maintained as project-specific references. These records provide evidence of the implementation and testing performed for the project.

11. RESULT AND OUTPUT

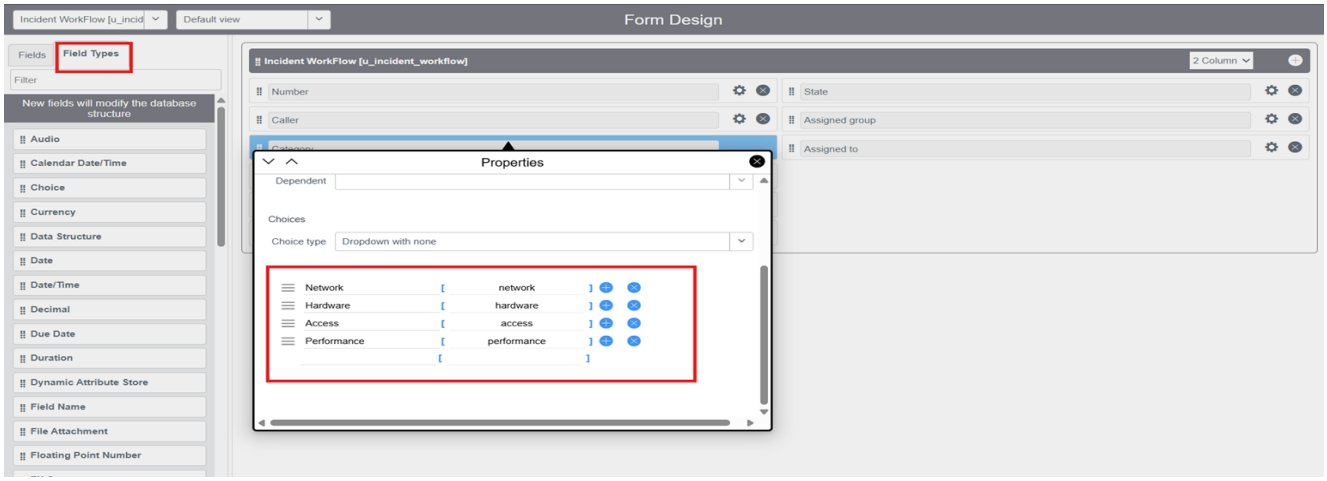
11.1 Servicenow Home Page



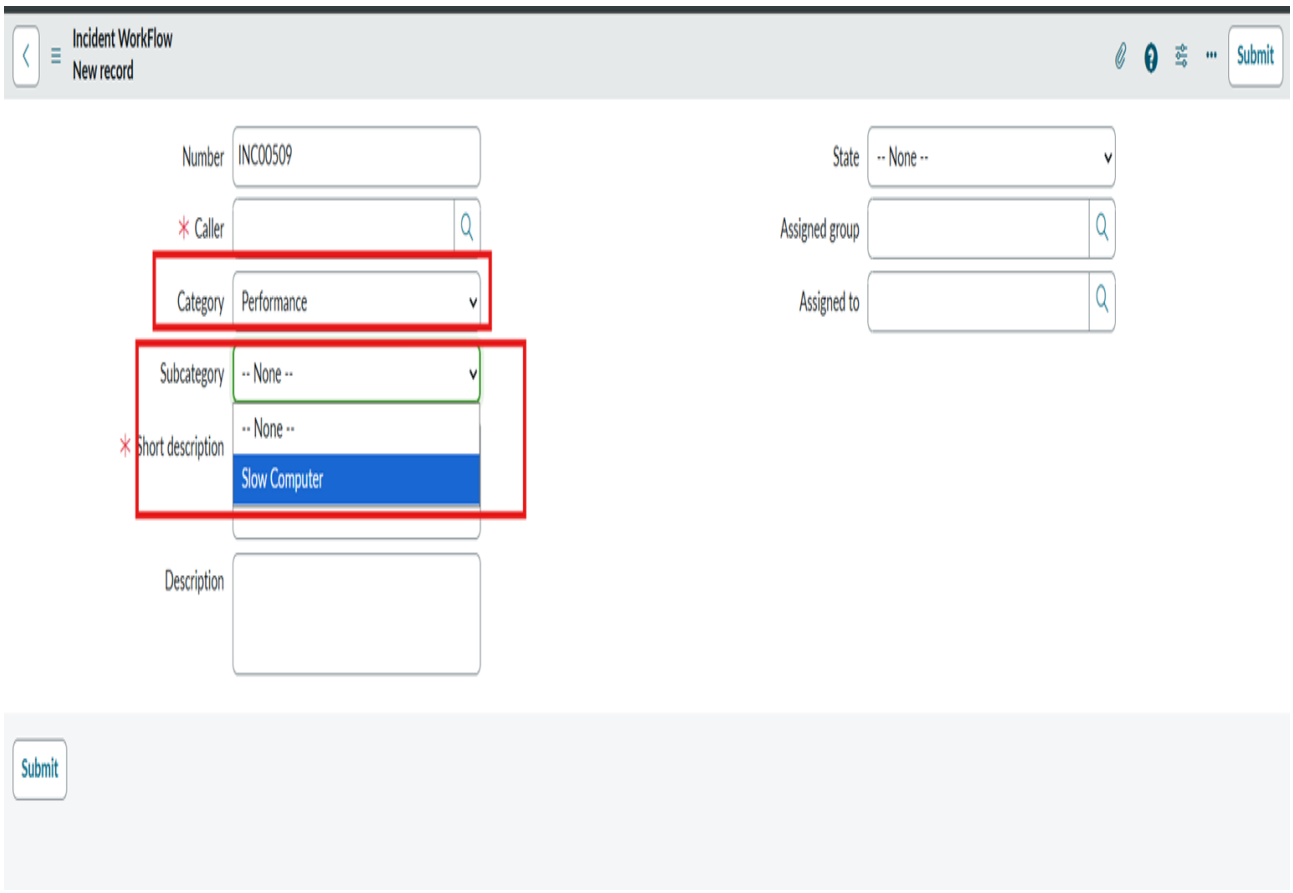
11.2 Servicenow Login Page



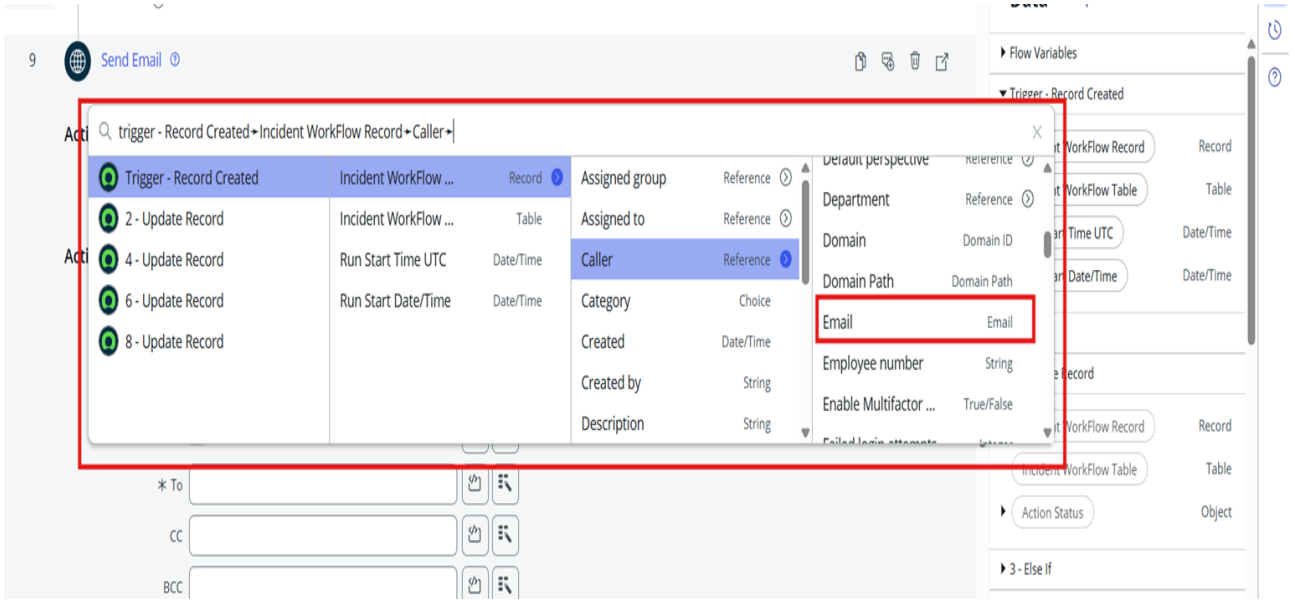
11.3 Form Design Output



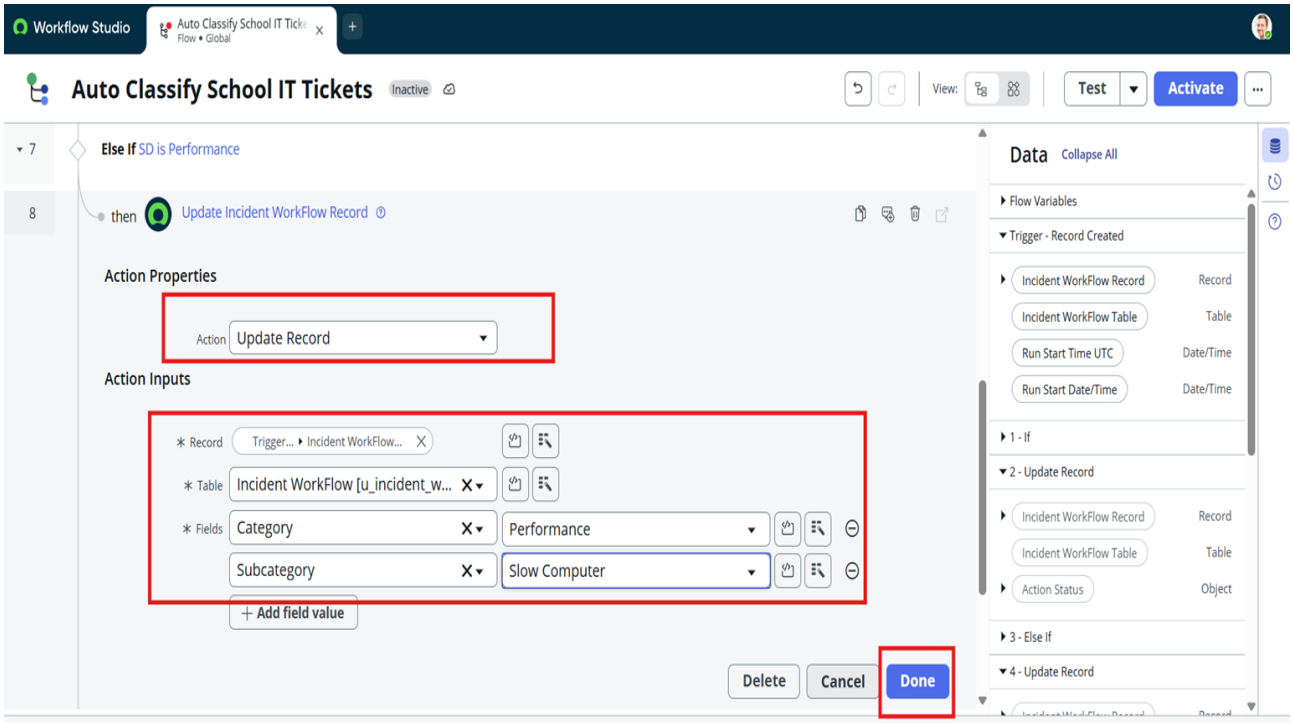
11.4 Backend Development Output



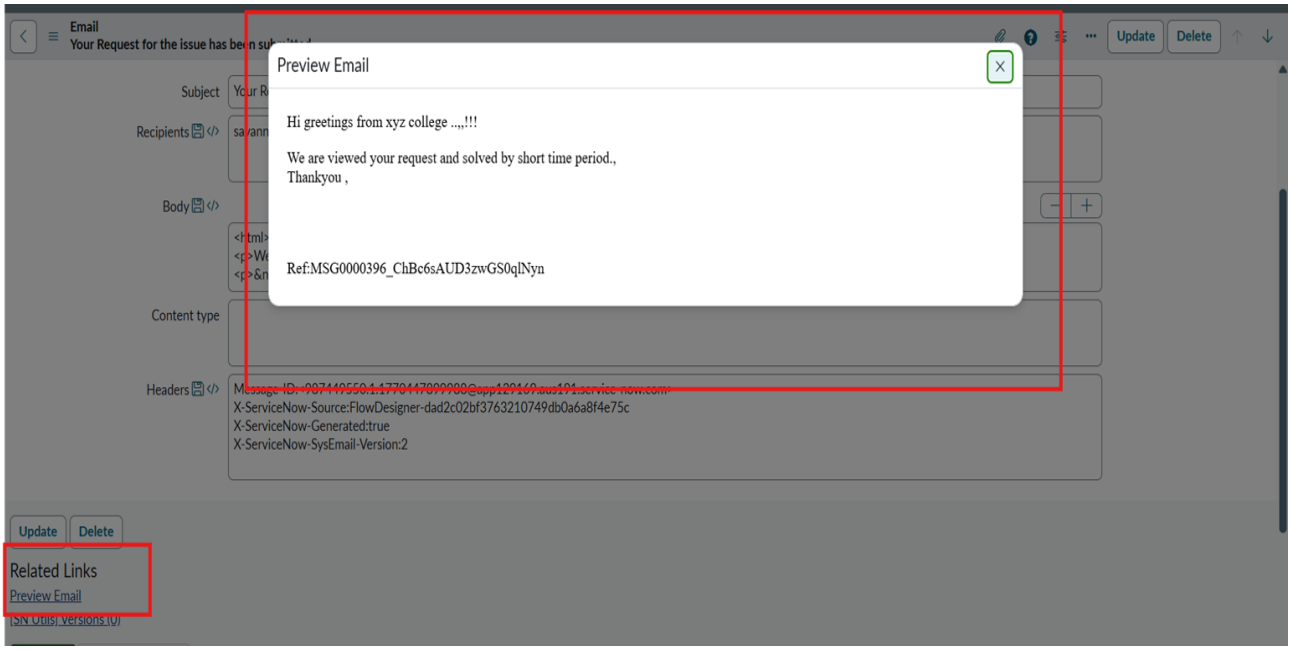
11.5 Email Notification



11.6 School IT Ticket Output



11.7 Validation & Security



11.8 Conclusion

